Math 472: Homework 07

Due Monday, March 23

Problem 1 (Exercise 9.8)**.** Let $Y_1, \dots, Y_n$ denote a random sample from a probability density function $f(y)$, which has unknown parameter θ. If $\widehat{\theta}$ is an unbiased estimator of θ, then under very general conditions

$$\mathrm{Var}(\widehat{\theta}) \geq \frac{1}{\mathcal{I}(\theta)}, \quad \text{where} \quad \mathcal{I}(\theta) = n\mathbb{E}\left[-\frac{\partial^2}{\partial\theta^2}\Big[\log(f(Y))\Big]\right].$$

(Note that in the definition of $\mathcal{I}(\theta)$, the Y is a random variable with density function f.) If $\mathrm{Var}(\widehat{\theta}) = \frac{1}{\mathcal{I}(\theta)}$, the estimator $\widehat{\theta}$ is said to be *efficient*.

Suppose that $p(y)$ is the Poisson probability function with mean $\lambda > 0$. Show that $\overline{Y}$ is an efficient estimator of λ.

Problem 2. Given a function $h : \mathbb{R} \to \mathbb{R}$, we say that $x^* \in \mathbb{R}$ is a ***maximizer*** of h if

$$h(x) \leq h(x^*) \text{ for all } x \in \mathbb{R}.$$

We say that $x^* \in \mathbb{R}$ is a ***local maximum*** of h if there exists some $\delta > 0$ such that

$$h(x) \leq h(x^*) \text{ for all } x \in (x^* - \delta, x^* + \delta).$$

We say that a function h is ***strictly increasing*** if

$$h(x) < h(y) \text{ whenever } x < y.$$

Let $f : \mathbb{R} \to \mathbb{R}$ be a function, and let $g = \phi \circ f$, where ϕ is a strictly increasing function whose domain contains the range of f.

(a) Show that x^* is a maximizer of f if and only if it is a maximizer of g. (You don't need to assume that f or ϕ is differentiable.)

(b) Show that x^* is a local maxium of f if and only if it is a local maximum of g. Conclude that f and g have the same local maxima. (Again, you don't need to assume that f or ϕ is differentiable).

(c) Does the same conclusion hold for local minima? *[Hint: consider the function $-f(x)$.]*

In our class, when finding the maximum likelihood estimator, we need to maximize the likelihood $L(\theta)$. The previous problem shows that it is enough to maximize $\log L(\theta)$, since $\phi(x) := \log(x)$ is strictly increasing. In fact, we can obtain stronger conclusions if we are allowed to assume that f and ϕ are differentiable, as shown in the next problem.

Problem 3. Let f, g, and ϕ be defined as in Problem 2. Further, assume that f and ϕ are both twice differentiable, and that $\phi'(x) > 0$ for all $x \in \mathbb{R}$.

(a) Show that $x^* \in \mathbb{R}$ is a critical point of f if and only if it is a critical point of g.

(b) Suppose that $x^* \in \mathbb{R}$ is a critical point of f and g. Show that $f''(x^*)$ and $g''(x^*)$ are either (i) both positive, (ii) both negative, or (iii) both zero.

(c) What additional information does part (b) tell you about functions f and g?

Problem 4. There is a subtlety to the assumptions used in Problems 2 and 3, which we highlight in this problem. Let $\phi : \mathbb{R} \to \mathbb{R}$ be a differentiable function.

(a) Show that if $\phi'(x) > 0$ for all $x \in \mathbb{R}$, then ϕ is strictly increasing. *[Hint: use the mean value theorem].*

(b) Show that the converse to part (a) does not hold (i.e., give a counterexample).

Problem 5. Write `R` code to replicate Figure 9.2 (page 449) in the textbook.

Problem 6 (Exercise 9.3 and 9.15?)**.** Let $Y_1, \ldots, Y_n$ denote a random sample from the uniform distribution on the interval $(\theta, \theta + 1)$. Let

$$\widehat{\theta}_1 = \overline{Y} - \frac{1}{2} \quad \text{and} \quad \widehat{\theta}_2 = \max(Y_1, \ldots, Y_n) - \frac{n}{n-1}.$$

(a) Show that both $\widehat{\theta}_1$ and $\widehat{\theta}_2$ are unbiased estimators of θ.

(b) Find the efficiency of $\hat{\theta}_1$ relative to $\widehat{\theta}_2$.

(c) Show that both $\hat{\theta}_1$ and $\widehat{\theta}_2$ are consistent estimators of θ.

Problem 7. Let X be a random variable with cdf

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan(x), \quad (x \in \mathbb{R})$$

(This is called a Cauchy random variable.)

(a) Show that $\mathbb{E}[X]$ is not defined.

(b) Let $Y_n = \frac{X_1 + \ldots + X_n}{n}$. For $n \in \{100, 10000, 100000\}$, sample a large number of independent replicates of Y_n, and plot the histograms. Explain what you see.

Problem 8 (Exercise 9.37)**.** Let $X_1, \ldots, X_n$ denote n independent and identically distributed Bernoulli random variables such that

$$\mathbb{P}[X_i = 1] = p \quad \text{and} \quad \mathbb{P}[X_i = 0] = 1 - p$$

for each $i = 1, \ldots, n$. Let $S_n = X_1 + \ldots + X_n$. Show that S_n is sufficient for p by using the factorization criterion.

Problem 9. Suppose $Y_1, \ldots, Y_n$ is a random sample from a geometric distribution with parameter p. Show that $\overline{Y}$ is sufficeient for p.

Problem 10. Let $Y_1, \ldots, Y_n$ be a random sample from a normal distribution with mean μ and variance σ^2.

(a) If μ is unknown and σ^2 is known, show that $\overline{Y}$ is sufficient for μ.

(b) If μ is known and σ^2 is unknown, show that $\sum_{i=1}^n (Y_i - \mu)^2$ is sufficient for σ^2.

Problem 11. Suppose $Y_1, \ldots, Y_n$ is a random sample from a Poisson distribution with mean $\lambda > 0$.

(a) Use the factorization criterion to find a sufficient statistic for λ.

(b) Find the MLE $\widehat{\lambda}$ of θ.

Problem 12. Suppose $Y_1, \ldots, Y_n$ is a random sample from a uniform distribution with pdf

$$f(y \mid \theta) = \frac{1}{2\theta + 1} \cdot \mathbf{1}_{[0 \le y \le 2\theta + 1]}.$$

Find the mle of θ.

Problem 13 (Optional)**.** Make a math meme related to statistics.